

Firstly, we run a loop as long as the game continues. It runs until the programme is ended by the command (System.end). Each iteration of this loop set up the game fresh. All the canvas and objects are constructed again. There exists player, arrow, bar and 11 ball objects that are constructed. The variables and constants are in environment file. There are other files that supplements us to retrieve information and methods to manipulate information.

Secondly, enter a loop that lasts until the game is won or lost. In each iteration of this inner loop, we first move the objects, then check if the objects cross, and then draw the object that satisfies conditions to the canvas. Cross of some objects causes actions other than the predefined movements of the objects, so we check if the borders of some objects cross or if their borders collide with predetermined intervals. First, we check.

We check if the balls cross with player object. There are 4 possible crossing defined. The ball can cross from on left or right side; from on top side; from on left upper corner; or from right upper corner. If the ball exists, it will cause assigning a value to a variable that keeps if the game is lost (this will cause not entering the next iteration of the loop, thus the end of the game if user does not want to play again). This is checked for every ball.

We check if the balls cross with arrow objects. There are 2 possible crossing defined. The ball can cross from left or right side; or cross with the point of the arrow. If the ball and arrow exists, it will cause the ball and arrow not to exist anymore. So, it will result the popped ball not to appear on the canvas in this and following iterations. This is checked for every ball. This result the arrow not to appear until it is refreshed by pressing space bar. Spacebar does not activate refreshment until the arrow disappears. This command is in player file.

We check if the objects exist before drawing and then draw them on the canvas by using the coordinates stored on the objects.

We draw the background first, then the arrow if it exists, then the floor bar, then the player, then one by one the balls if they exist, then the time bar.

We change the color of the time bar. The bar goes from yellow to red. We gradually (proportional to time past) decrease the green light in the yellow light and get redder light until fully red light. The bar moves left thus shrinks until the game ends and is not visible after the game ends after 40 seconds.

Game may end if the time is up, if the ball crosses the player and if all the balls are cleaned/popped. We check if game is lost and draw game screen telling user has lost, we check if it is won and draw a game screen telling user has won.

Lastly, check if the user wants to play again or end the program by asking him/her to press Y or N for Yes or No. User can refresh the whole game by pressing Y or exit the program by pressing N.

[Screen Recording 2023-04-17 at 02.01.31.mov](#)

Note: Environment provided a class that we can retrieve and manipulate variables. You can check the java file to see all the classes.

